

Self-Collision Models for ITER-EDA RF Case 1: fixed-temperature (`isc=1`) vs. adaptive (`isc=2`) Maxwellian background vs. the nonlinear operator (`isc=-1`)

FP2D_QLRF_NL solver — post-processing analysis

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Revision history. v1.0 — initial three-way comparison, $D_{\perp\perp}/F_{\perp}$ diagnostics and mechanism.
v1.1 — adds the carried-out diagnostic tests (Section 6: common- f operator comparison, velocity-space SC power density, and the P_{SC} time traces with the self-collision timescale), and refines the discussion of the fixed-temperature model.

1 Case Setup

This is a *pure ICRF* case: there is no particle source (`isource=0`). The simulation evolves a tritium minority ($A = 3$, $Z = 1$) that starts from a Maxwellian at $T = 12$ keV (thermal equilibrium with the bulk) and is heated at the 2nd cyclotron harmonic (60.9 MHz; $f_c = 30.5$ MHz). The background is a deuterium plasma at $T = 12$ keV and $n_e = 1.5 \times 10^{20} \text{ m}^{-3}$; the tritium fraction is $x_T = 0.5$ ($n_T = 7.5 \times 10^{19} \text{ m}^{-3}$). The velocity-space grid is 150×150 in $(v_\perp, v_\parallel)$, with $v_\perp^{\text{max}} = 3.5 \times 10^7 \text{ m s}^{-1}$ and $v_\parallel^{\text{max}} = 4 \times 10^6 \text{ m s}^{-1}$.

The three runs use identical solver parameters — same grid, same two-phase time stepping (`ntimes` = 600, 2000; `timestep` = 1 ms, 25 ms), same initial condition (`iold`=0, `istart`=1, i.e. the $T = 12$ keV Maxwellian), and same convergence tolerance (10^{-4}). They differ *only* in the self-collision flag `isc`. Any difference in the results is therefore attributable solely to the self-collision model, not to the numerical settings.

2 Self-Collision Models

2.1 Nonlinear operator (`isc=-1`)

The fully nonlinear operator uses the actual distribution f as both the test-particle and the field (background) distribution:

$$C[f, f] = \nabla_v \cdot [\mathbf{A}(f) f - \mathbf{D}(f) \cdot \nabla_v f], \quad (1)$$

where the friction vector $\mathbf{A}$ and diffusion tensor $\mathbf{D}$ are functionals of f through the Rosenbluth potentials. This operator satisfies *exact energy conservation*:

$$\int C[f, f] \frac{1}{2} m v^2 d^3 v \equiv 0 \quad \forall f. \quad (2)$$

2.2 Linearised operators (`isc=1` and `isc=2`)

Both Maxwellian-background models replace the field distribution by an isotropic Maxwellian $f_M(T_b)$ and use the *same* linearised operator $C[f, f_M(T_b)]$, computed analytically (Chandrasekhar function) in `cblin`. They differ *only* in the choice of the background temperature T_b :

$$\text{isc}=1: \quad T_b = T_{\text{Stix}} = 12 \text{ keV (fixed, the bulk temperature)}, \quad (3)$$

$$\text{isc}=2: \quad T_b = T_{\text{eff}}(t) = \frac{2T_\perp + T_\parallel}{3} \quad (\text{evolving with the minority's own energy}). \quad (4)$$

The Maxwellian background is

$$f_M(v; T_b) = \frac{n}{\pi^{3/2} v_{\text{th}}^3} \exp\left(-\frac{v^2}{v_{\text{th}}^2}\right), \quad v_{\text{th}} = 9.79 \times 10^3 \sqrt{T_b[\text{eV}]/A}. \quad (5)$$

A key point for what follows: *neither* linearised operator conserves energy exactly when $f \neq f_M(T_b)$, because the background acts as an external reservoir of infinite heat capacity. The energy moment

$$P_{\text{SC}} = \int C[f, f_M(T_b)] \frac{1}{2} m v^2 d^3 v \quad (6)$$

is the net collisional energy transfer between f and that reservoir. Its sign and magnitude depend critically on T_b , as the results below show.

Table 1: Final-state diagnostics for the three runs (identical solver settings). Powers in MW m^{-3} .

Quantity	isc=1 (NLMax1) fixed $T_b = 12 \text{ keV}$	isc=2 (NLMax2) adaptive $T_b = T_{\text{eff}}$	isc=-1 (NLSC) exact
Converged?	Yes (step 945)	Yes (step 1374)	No (plateau) ¹
T_{eff} (keV)	19.4	78.9	22.2
$\langle E_{\text{kin}} \rangle$ (keV)	29.1	118.3	33.3
$E_{\perp}$ (keV)	22.8	105.3	26.6
$E_{\parallel}$ (keV)	6.34	13.0	6.71
Anisotropy	64.2%	80.1%	66.5%
P_{RF}	1.128	3.413	1.247
$P_{\text{coll,e}}$	-0.271	-2.430	-0.372
$P_{\text{coll,D}}$	-0.477	-4.486	-0.872
$P_{\text{SC},\perp}$	-0.394	+2.131	-0.137
$P_{\text{SC},\parallel}$	+0.014	+1.359	+0.135
$P_{\text{SC,total}}$	-0.380	+3.490	-0.002
Power balance residual	-2.2×10^{-4}	-1.3×10^{-2}	-1.6×10^{-4}

3 Results

3.1 Final-state summary

3.2 The central observation

isc=1 tracks the exact operator (**isc=-1**) closely *at steady state*: $T_{\text{eff}} = 19.4$ vs. 22.2 keV , $P_{\text{RF}} = 1.13$ vs. 1.25 MW m^{-3} , anisotropy 64 vs. 67%. **isc=2**, by contrast, is wildly off: $T_{\text{eff}} = 79 \text{ keV}$ (a factor ~ 3.5 too hot) and P_{RF} a factor ~ 3 too large. (As Test 1 in Section 6 shows, this steady-state agreement does *not* mean the **isc=1** operator is locally accurate — only that its error has a benign sign.)

This is the result that demands explanation, because **isc=1** and **isc=2** use the *same* linearised operator and differ only in T_b . The failure of **isc=2** therefore *cannot* be attributed to the linearisation itself, nor to the fact that f is non-Maxwellian (both models see the same non-Maxwellian f). The culprit must be the choice $T_b = T_{\text{eff}}$.

The time evolution of T_{eff} (Fig. 1) makes the contrast vivid. **isc=1** and **isc=-1** both rise from 12 keV and level off near $19\text{--}22 \text{ keV}$ within $\sim 2 \text{ s}$, whereas **isc=2** climbs monotonically through these values without arrest, passing 77 keV and still rising at the right edge of the plot.

3.3 Self-collision power: the sign tells the story

The energy moment P_{SC} , Eq. (6), behaves qualitatively differently in the three cases:

isc=-1: $P_{\text{SC}} \approx -0.002 \text{ MW m}^{-3}$ (≈ 0). Energy-conserving to numerical precision. The split $P_{\text{SC},\perp} = -0.137$, $P_{\text{SC},\parallel} = +0.135$ shows pure redistribution (pitch-angle scattering / isotropisation) with no net energy change.

isc=1: $P_{\text{SC}} = -0.380 \text{ MW m}^{-3}$ (sink). A modest *negative* (cooling) term. The cold 12 keV background drags the RF-heated minority back toward the bulk. This opposes the RF drive — it is *stabilising* — and the steady state settles close to the true one.

¹The **isc=-1** run does not formally converge: dE/E plateaus at $\approx 4.2 \times 10^{-4}$, just above the 10^{-4} tolerance, while the power residual $dP/P_d \sim 10^{-7}$ is fully converged. This is a numerical floor of the energy-norm estimator for the nonlinear operator, not ongoing physical evolution: T_{eff} is stationary to within 0.1 keV over the last 1000 steps.

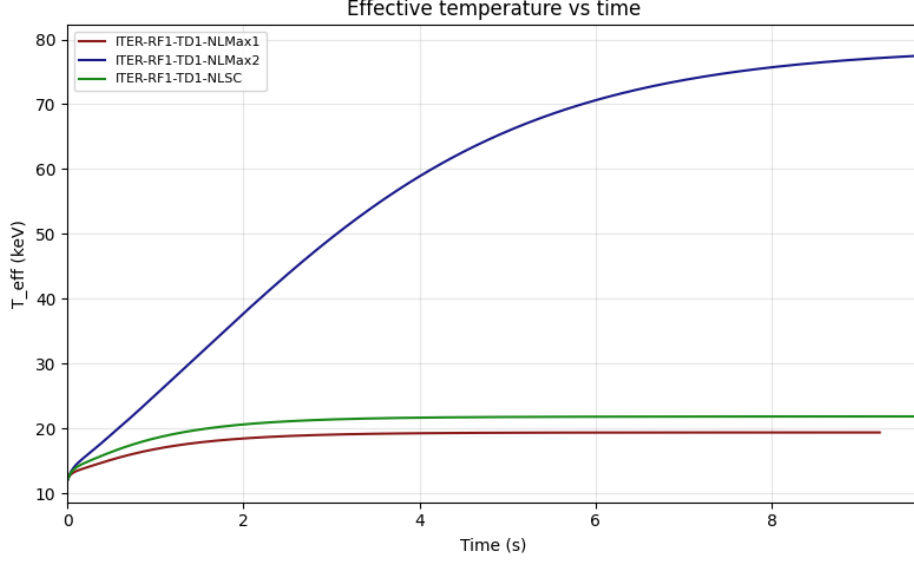


Figure 1: Effective temperature $T_{\text{eff}}(t)$ for the three runs. **isc=1** (NLMax1, red) and **isc=-1** (NLSC, green) saturate close together at ≈ 19 and 22 keV; **isc=2** (NLMax2, blue) runs away monotonically, reaching ≈ 79 keV at its own (later) convergence. The horizontal axis is truncated at the **isc=1** convergence time ($t \approx 9.2$ s), where the **isc=2** curve is still climbing.

isc=2: $P_{\text{SC}} = +3.490 \text{ MW m}^{-3}$ (**source**). A large *positive* (heating) term, comparable to the entire RF input. The self-collision operator behaves as an energy source. This is *destabilising* and drives the runaway.

4 Diffusion and friction diagnostics

To pinpoint the mechanism we output, at the end of each run, the perpendicular diffusion coefficient $D_{\perp\perp}$ and the perpendicular friction $F_{\perp}$ of the self-collision operator along the $v_{\parallel} = 0$ slice. For **isc=1,2** these come directly from **cbilin** ($D_{\perp\perp} = \Theta + v_{\parallel}^2 \Phi$, $F_{\perp} = -v_{\perp} \Psi$); for **isc=-1** they are reconstructed from the numerically-computed Rosenbluth potentials ($D_{\perp\perp} = -\frac{4\pi\gamma}{n} \partial^2 \psi / \partial v_{\perp}^2$, $F_{\perp} = -\frac{4\pi\gamma}{n} \partial \phi / \partial v_{\perp}$).

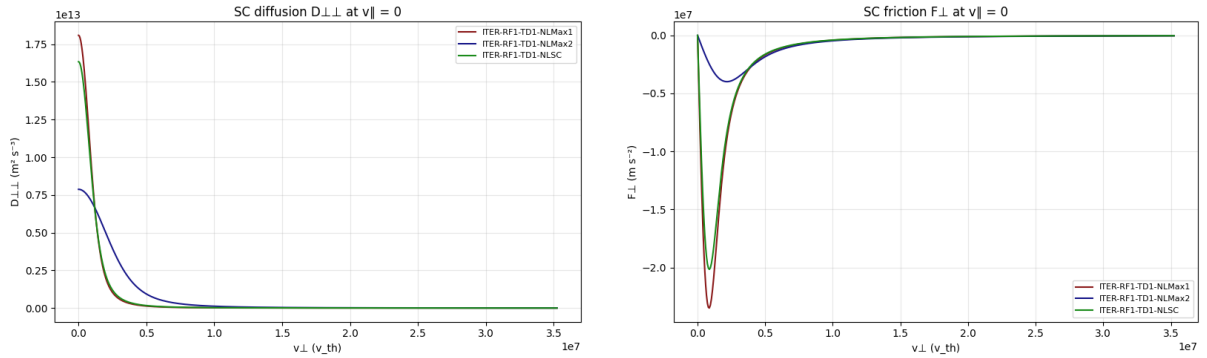


Figure 2: Self-collision perpendicular diffusion $D_{\perp\perp}$ (left) and friction $F_{\perp}$ (right) at $v_{\parallel} = 0$, for the final state of each run. **isc=1** (red) and **isc=-1** (green) nearly coincide; **isc=2** (blue) is qualitatively different: its diffusion is lower at the axis but far broader in $v_{\perp}$, and its friction is $\sim 5\times$ weaker with its minimum displaced to higher $v_{\perp}$.

The diagnostics (Fig. 2) are unambiguous:

- **isc=1 and isc=-1 coincide.** Both $D_{\perp\perp}$ (peak $\approx 1.7\text{--}1.8 \times 10^{13} \text{ m}^2 \text{ s}^{-3}$ at the axis, decaying to zero by $v_{\perp} \approx 5 \text{ Mm s}^{-1}$) and $F_{\perp}$ (deep well, minimum $\approx -2 \times 10^7 \text{ m s}^{-2}$ at $v_{\perp} \approx 0.8 \text{ Mm s}^{-1}$) are essentially the same for the fixed-12 keV background and the exact operator.
- **isc=2 is qualitatively wrong.** Its diffusion peak is less than half ($\approx 0.8 \times 10^{13}$) but *extends far out in $v_{\perp}$* — significant diffusion persists out to $v_{\perp} \approx 10 \text{ Mm s}^{-1}$, where the other two have vanished. Its friction well is $\sim 5\times$ shallower ($\approx -0.4 \times 10^7$) and displaced outward to $v_{\perp} \approx 2 \text{ Mm s}^{-1}$.

5 Physical Interpretation

5.1 Density-weighted field vs. energy-weighted temperature

The friction and diffusion felt by a test particle are *number-density-weighted* integrals over the field particles (the Rosenbluth potentials are convolutions of f with $|v - v'|$ and $1/|v - v'|$). They are therefore dominated by the *dense, cold bulk* of the distribution, not by the sparse high-energy tail. The bulk of the minority, even after RF heating, remains close to the 12 keV core, because the RF populates a thin perpendicular tail that carries much energy but little particle number.

This is precisely why **isc=1** works (Fig. 2): a fixed Maxwellian at the *bulk* temperature (12 keV) is an excellent proxy for the true collisional field, because the true field is set by the bulk. The fixed-background operator is, in effect, an extra drag against a cold thermal population — physically equivalent to an additional bulk-ion collision channel, with the stabilising (cooling) sign. This agreement concerns the $D_{\perp\perp}/F_{\perp}$ *shapes* near the bulk and the self-consistent steady state; applied to a *fixed* f the energy moment of the **isc=1** operator still differs markedly from the exact one (it is a sizeable sink, not ≈ 0) — see Test 1, Section 6.

$T_{\text{eff}} = (2T_{\perp} + T_{\parallel})/3$, on the other hand, is an *energy-weighted* moment. It is pulled up to 79 keV by the perpendicular RF tail. Using T_{eff} as the background temperature (**isc=2**) therefore characterises the collisional field by the hot tail rather than the cold bulk — the wrong weighting entirely. The background Maxwellian is made $\sim 6\times$ too hot, so $v_{\text{th}}(T_b)$ is $\sim 2.5\times$ too large, which directly produces the broad diffusion and weak friction seen in Fig. 2.

5.2 Why isc=2 becomes an energy source

The equilibrium of a linearised operator is $f = f_M(T_b)$, enforced by the fluctuation–dissipation balance between friction and diffusion, $F_{\perp}/D_{\perp\perp} \sim v_{\perp}/v_{\text{th}}^2(T_b)$. Lowering this ratio (hotter background) means the operator relaxes f toward a hotter target. Concretely, in **isc=2**:

- the broad $D_{\perp\perp}$ keeps up-scattering particles in the perpendicular tail (injecting $\perp$ energy) far out in $v_{\perp}$, where it should have died away;
- the friction $F_{\perp}$, suppressed $\sim 5\times$, is far too weak to pull those particles back.

The net perpendicular energy moment is therefore strongly positive ($P_{\text{SC},\perp} = +2.13 \text{ MW m}^{-3}$), and the operator acts as a heating source rather than a relaxation term. Because $T_b = T_{\text{eff}}$ is slaved to the minority’s own energy, this closes a positive feedback loop:

$$T_{\text{eff}} \uparrow \longrightarrow T_b = T_{\text{eff}} \uparrow \longrightarrow F_{\perp}/D_{\perp\perp} \downarrow \longrightarrow P_{\text{SC}} \uparrow \longrightarrow T_{\text{eff}} \uparrow\uparrow.$$

There is no restoring force. T_{eff} runs up until the genuine Coulomb losses to electrons and bulk D ions — which scale with T_{eff} and do *not* share this feedback — grow large enough to balance the spurious source, which happens only at $T_{\text{eff}} \approx 79 \text{ keV}$.

In **isc=1** the same fluctuation–dissipation argument runs the other way: the fixed cold background gives strong friction and narrow diffusion, so the operator relaxes f toward the 12 keV bulk, removing energy ($P_{\text{SC}} < 0$). This is a stable negative feedback and the steady state stays close to the true one.

5.3 Why P_{RF} tracks T_{eff}

The inflated P_{RF} in **isc=2** is a *secondary* consequence, not an independent effect. The 2nd-harmonic quasilinear absorption scales with the number of resonant ions and the Bessel overlap $J_1^2(k_{\perp}v_{\perp}/\Omega_c)$. The artificially hot, broad perpendicular distribution places far more ions in resonance, so P_{RF} grows with T_{eff} and feeds the same runaway. The factor-3 excess in P_{RF} is thus downstream of the factor-3.5 excess in T_{eff} .

6 Diagnostic tests

Three of the diagnostics suggested in the previous version have been carried out. They confirm the mechanism and sharpen one point about the fixed-temperature model.

6.1 Test 1 — the three operators on a common f

Evaluating all three self-collision operators on the *same* distribution (the converged **isc=-1** VDF, $T_{\text{eff}} \approx 22$ keV) removes the confound of the different final states (Table 2).

Table 2: Self-collision power (MW m^{-3}) of each operator applied to the single common distribution f (the converged **isc=-1** VDF).

Operator on common f	$P_{\text{SC},\perp}$	$P_{\text{SC},\parallel}$	$P_{\text{SC},\text{total}}$
isc=-1 (exact)	−0.138	+0.136	≈ 0
isc=1 ($T_b = 12$ keV)	−0.624	−0.077	−0.70
isc=2 ($T_b = T_{\text{eff}} = 22$ keV)	+1.70	+1.38	+3.08

The exact operator conserves energy (≈ 0) and merely transfers $\perp \rightarrow \parallel$ (isotropisation of the RF-driven anisotropy). Both linearised operators, applied to this *same* f , are far off: **isc=1** is a large *sink* (−0.70) and **isc=2** a large *source* (+3.08).

This is the key refinement of v1.1. The good steady-state agreement of **isc=1** (Table 1) is *not* because the fixed-temperature operator is locally accurate — on a fixed f it is not. It is because its error has the *stabilising* sign (a drag/sink): the run self-adjusts to a slightly cool but stable equilibrium where RF input balances the over-strong collisional loss. **isc=2**’s error has the opposite, *destabilising* sign (a source), for which no stable balance exists — hence the runaway. The opposite signs follow directly from the background temperature relative to the distribution: f has $T_{\text{eff}} = 22$ keV (energy-weighted, inflated by the perpendicular tail) but a particle bulk still near 12 keV, so the 12 keV bath of **isc=1** cools the tail while the 22 keV bath of **isc=2** heats the bulk.

6.2 Test 2 — velocity-space density of the SC power

Figure 3 maps $dP_{\text{SC}}/d^3v = \frac{1}{2}mv^2 C_{\text{SC}}[f]$ (source > 0 , sink < 0) for each run; Fig. 4 compares the $v_{\parallel} = 0$ slices.

Why **isc=2 becomes a *perpendicular* source.** A collisional operator drives the local logarithmic slope of f toward that of its background Maxwellian (the fluctuation–dissipation balance

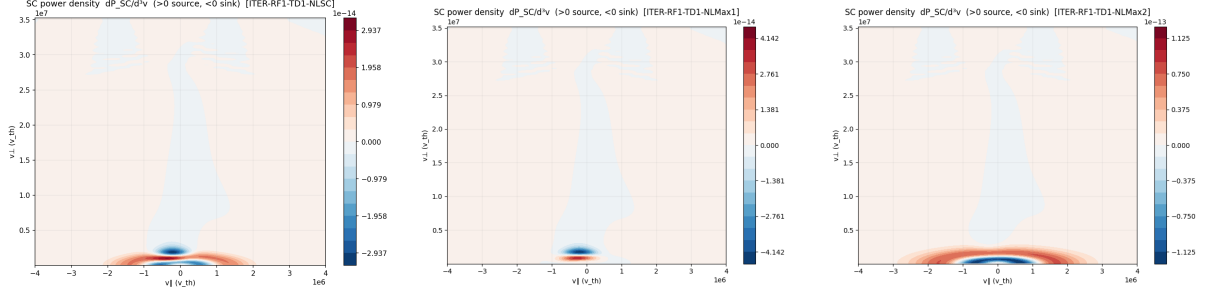


Figure 3: SC power density dP_{SC}/d^3v for $\text{isc}=-1$ (left), $\text{isc}=1$ (centre) and $\text{isc}=2$ (right). $\text{isc}=-1$ and $\text{isc}=1$ are tightly localised at the bulk (a dipolar pitch-angle pattern; net ≈ 0 and net sink respectively). $\text{isc}=2$ shows a broad *source* band spread in $v_{\parallel}$ and reaching up in $v_{\perp}$, with a sink core at $v_{\parallel} \approx 0$.

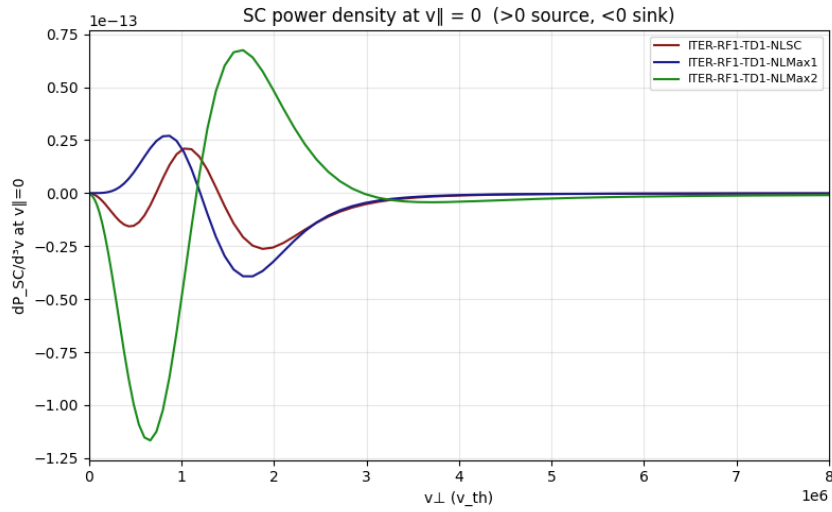


Figure 4: dP_{SC}/d^3v at $v_{\parallel} = 0$. $\text{isc}=2$ (green) has a large positive lobe at $v_{\perp} \approx 1.6 \text{ Mm s}^{-1}$ — a perpendicular *source* in the bulk — absent in $\text{isc}=-1$ (red) and $\text{isc}=1$ (blue).

$D \partial_v f + F f \rightarrow 0$, i.e. $\partial_v \ln f \rightarrow -2v/v_{\text{th}}^2(T_b)$. Where f is *colder* than T_b (steeper than $f_M(T_b)$) diffusion beats friction and particles are scattered *up* in energy (source); where f is hotter than T_b , friction wins (sink). For $\text{isc}=2$, $T_b = T_{\text{eff}} = 22 \text{ keV}$ sits *above* the bulk core ($\sim 12 \text{ keV}$), so across the densely populated bulk f is colder than the background and the operator flattens it, pumping particles up in energy. Because $D_{\perp\perp}$, set by the hot broad background, is the dominant far-reaching term and the cylindrical phase space ($2\pi v_{\perp}$) favours $v_{\perp}$, the up-scattering deposits energy preferentially in the perpendicular direction. In short, $\text{isc}=2$ applies a 22 keV heat bath to a 12 keV bulk: the “self”-collision operator heats the bulk rather than relaxing it — exactly the $+3 \text{ MW m}^{-3}$ non-conservation — and it is perpendicular because the diffusion and the phase-space weighting favour $v_{\perp}$.

6.3 Test 5 — time traces and the self-collision timescale

Figure 5 overlays $P_{\text{SC},\perp}(t)$ and $P_{\text{SC},\parallel}(t)$ for the three runs. For $\text{isc}=2$, $P_{\text{SC},\perp}$ starts slightly *negative*, crosses zero at $t \approx 0.04 \text{ s}$, and then grows positive.

The sign flip is understood as follows. At $t = 0$ the distribution is the 12 keV Maxwellian, so $T_b \approx T_{\text{eff}} \approx$ the bulk temperature and $f \approx f_M(T_b)$: the operator is nearly null. As RF heats $v_{\perp}$, f first becomes anisotropic while T_{eff} is still near the bulk, and the operator does ordinary

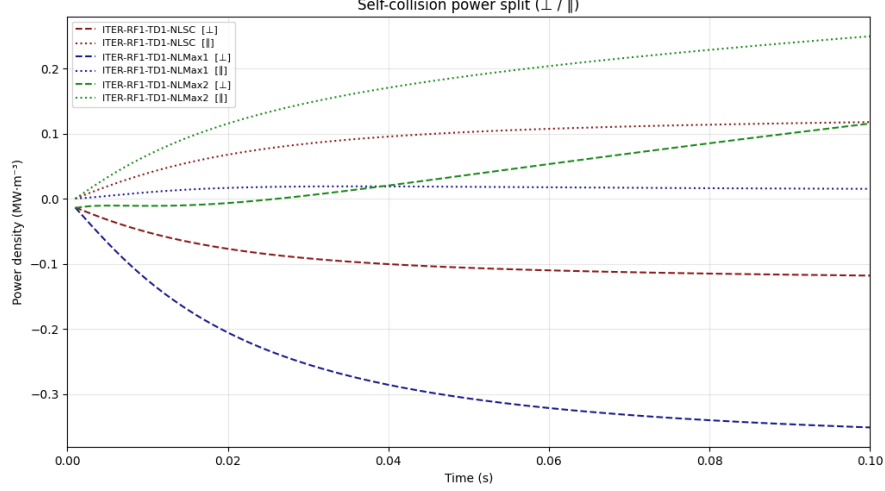


Figure 5: Self-collision power split $P_{\text{SC},\perp}$ (dashed) and $P_{\text{SC},\parallel}$ (dotted) vs time. For $\text{isc}=2$ (green) the perpendicular component flips from sink to source at $t \approx 0.04\text{ s}$.

isotropisation ($\perp \rightarrow \parallel$): hence $P_{\text{SC},\perp} < 0$ and $P_{\text{SC},\parallel} > 0$ initially, like the exact operator. As T_{eff} climbs above the bulk core, the spurious heating of Test 2 switches on and grows, adding a positive contribution to both channels; in $\perp$ it eventually overtakes the isotropisation sink and $P_{\text{SC},\perp}$ crosses zero.

The crossover time is set by the *ion self-collision time*. For the tritium minority at 12–13.5 keV, $n_T = 7.5 \times 10^{19} \text{ m}^{-3}$, $\ln \Lambda \approx 21.5$ (NRL formulary), $\tau_{ii} \approx 29\text{--}35\text{ ms}$ — right on the observed $\sim 40\text{ ms}$. (The RF time to raise T_{eff} by the $\sim 1.5\text{ keV}$ needed to flip is comparable, $\sim 30\text{ ms}$, as it must be at the crossover.) This gives a clean two-timescale picture: a *fast* clock, $\tau_{ii} \approx 30\text{--}40\text{ ms}$, on which self-collisions respond and the $\text{isc}=2$ operator’s character flips from sink to source; and a *slow* one, $\sim \text{seconds}$, on which the net energy balance drives the runaway to 79 keV (set by the much weaker electron drag plus the feedback). The pathology therefore announces itself within one self-collision time and then drives the slow runaway.

7 Further diagnostics not yet carried out

Tests 1, 2 and 5 above have been carried out (Section 6). The following would further strengthen the analysis:

- **Two background temperatures side by side.** Output both T_{eff} (energy-weighted) and a density-characteristic temperature T_n (e.g. from the curvature of $\ln f$ near the core, or a low-order moment dominated by the bulk). The gap T_{eff}/T_n quantifies exactly how far the $\text{isc}=2$ background departs from the collisionally-relevant temperature, and would let one test an “ $\text{isc}=2$ with $T_b = T_n$ ” variant.
- **Parallel and cross coefficients.** Add $D_{\parallel\parallel}$, $F_{\parallel}$ and the off-diagonal $D_{\perp\parallel}$ (slices and 2D maps) to confirm the anisotropic structure and check that the perpendicular channel is indeed the dominant error.
- **The Rosenbluth potentials ψ , ϕ** (or their key $v_{\perp}$ derivatives) for the exact field vs. the two Maxwellian backgrounds, to visualise the root cause one level below D and F .

8 Conclusions

1. **isc=1 tracks the exact operator at steady state; isc=2 does not.** With identical solver settings, **isc=1** gives $T_{\text{eff}} = 19.4 \text{ keV}$ and $P_{\text{RF}} = 1.13 \text{ MW m}^{-3}$, close to the exact **isc=-1** values (22.2 keV , 1.25 MW m^{-3}), whereas **isc=2** gives 79 keV and 3.41 MW m^{-3} .
2. **The failure is the background temperature, not the linearisation.** **isc=1** and **isc=2** use the same linearised operator on the same non-Maxwellian f ; only T_b differs. Neither conserves energy exactly, but with T_b at the bulk temperature the residual is a small stabilising sink, whereas with $T_b = T_{\text{eff}}$ it is a large destabilising source ($P_{\text{SC}} = +3.49 \text{ MW m}^{-3}$).
3. **Mechanism (from the D/F diagnostics).** The collisional field is density-weighted and dominated by the cold bulk, so a bulk-temperature background reproduces the true friction and diffusion (Fig. 2). T_{eff} is energy-weighted and inflated by the perpendicular RF tail; using it makes the background too hot, giving broad diffusion and $\sim 5\times$ too-weak friction. The operator then pumps perpendicular energy instead of relaxing it, and the $T_b = T_{\text{eff}}$ slaving closes a runaway feedback loop.
4. **The inflated P_{RF} is secondary,** driven by the artificially hot, broad perpendicular distribution improving the 2nd-harmonic resonant coupling.
5. **Recommendation.** **isc=2** as currently defined ($T_b = T_{\text{eff}}$) should not be used for RF-driven or otherwise non-Maxwellian distributions. If a self-consistent self-collision model cheaper than the full nonlinear operator is needed, **isc=1** (fixed bulk-temperature background) is far more reliable here. A possible improvement to **isc=2** would be to define its background temperature from a density-weighted measure T_n rather than the energy-weighted T_{eff} (diagnostic 3 above).